

Replication Package

Early Socio-Emotional Skills and Adolescent Offending: Evidence from Administrative Data*

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Overview

This replication package contains all code required to reproduce the data cleaning, factor analysis, and regression results reported in the paper. The analysis uses linked administrative data from the Department for Education (DfE) and the Ministry of Justice (MoJ), accessed via the Office for National Statistics (ONS) Secure Research Service. Raw data files are not included in the package due to data governance restrictions. Researchers seeking access should apply via the [Data Access](#).

The package is organised into three subfolders containing scripts executed sequentially. All file paths are set to placeholder `file_path` and must be updated to absolute paths before running. Stata scripts were written for Stata 17; R scripts were written for R 4.3.

Data Sources

Dataset	Description
Spring School Census	Pupil-level demographic, SES, school and SEND variables. Annual waves, Spring Terms.
EYFSP	Early Years Foundation Stage Profile scores at school reception, academic years 2006/07–2008/09. Three cohorts.
National Pupil Database (NPD)	KS2 attainment, school exclusions, absenteeism.
Police National Computer (PNC)	Individual-level criminal records linked to the NPD sample. Offending outcomes up to age 17.
Children Looked After (CLA)	Annual looked-after children records.
Children in Need (CIN)	Child protection and referral records.

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Software Requirements

Software	Version	Non-standard packages
Stata	17	reghdfe, mdesc, outreg2, polychoric, pcamat, winsor2, coefplot, estout, fre
R	4.3	lavaan, psych, GPArotation, semTools, sirt, haven, polycor, ggplot2, corrplot, mnormt, rlang, dplyr

Package Structure

The repository root (SocioEmotional_Offending/) contains three subfolders. Scripts within each subfolder are numbered and should be run in order. Intermediate .dta files produced by earlier scripts are inputs to later ones.

```

SocioEmotional_Offending/
├── clean_data/
│   ├── 01_eyfsp_clean.do
│   ├── 02_census_clean.do
│   ├── 03_PNC_clean.do
│   ├── 04_ks2_clean.do
│   ├── 05_absences_clean.do
│   ├── 06_exclusions_clean.do
│   ├── 07_cla_cin_clean.do
│   └── 08_merge_clean.do
├── factorization/
│   ├── 01_eyfsp_inspect.do
│   ├── 02_eyfsp_efa.do
│   ├── 03_esem_eyfsp.R
│   ├── 04_CFA_bifactor.R
│   └── 05_PCA_NPD.do
└── analysis/
    ├── 01_descriptives.do
    ├── 02_eyfsp_differences.do
    ├── 03_regressions.do
    ├── 04_ipw.do
    ├── 05a_main_mediation.do
    └── 05b_mediation_boot.do

```

Script Descriptions

clean_data/

File	Purpose
01_eyfsp_clean.do	Imports and cleans the EYFSP data. Converts string scores to numeric. Output: <code>eyfsp_clean.dta</code> .
02_census_clean.do	Imports and cleans the Spring School Census. Extracts demographic, SES, school and SEND variables; merges annual waves into a single wide panel; constructs cohort-aligned time-varying school. Output: <code>census_clean.dta</code> .
03_PNC_clean.do	Imports and cleans Police National Computer offending records. Constructs offence-level variables (offence type); retains primary guilty offences; collapses to one observation per offender. Output: <code>crime_clean.dta</code> .
04_ks2_clean.do	Imports and cleans KS2 examination results. Output: <code>ks2_clean.dta</code> .
05_absences_clean.do	Imports and cleans pupil-level absence records. Sums authorised, unauthorised, and exclusion-related absences across terms; constructs cohort-aligned cumulative absence measures from Reception to KS2. Output: <code>absences_clean.dta</code> .
06_exclusions_clean.do	Imports and cleans exclusion records. Classifies exclusion reasons and categories; constructs cohort-aligned cumulative ever-excluded indicators. Output: <code>exclusions_clean.dta</code> .
07_cla_cin_clean.do	Imports and cleans Children Looked After (CLA) and Children in Need (CIN) records. Constructs cohort-aligned indicators for care history, placement, category of need, and child protection plans. Outputs: <code>cla_clean.dta</code> , <code>cin_clean.dta</code> .
08_merge_clean.do	Merges all cleaned data sources into a single cross-sectional analysis file. Constructs derived variables for offending, exclusions, CIN/CLA status, and missing data indicators. Output: <code>clean_data.dta</code> .

factorization/

File	Purpose
01_eyfsp_inspect.do	Inspects internal consistency of EYFSP point scales. Identifies pairs of items with tetrachoric correlations above 0.90 and 0.85 for item reduction prior to factor analysis.
02_eyfsp_efa.do	Exploratory Factor Analysis (EFA) of EYFSP binary items using tetrachoric correlations and oblique quartimin rotation. Yields a two-factor solution used to define the ESEM anchor structure.

File	Purpose
03_esem_eyfsp.R	Estimates the Exploratory Structural Equation Model (ESEM) using the two-factor solution from the EFA. Constructs ESEM model syntax from EFA loadings with anchor constraints; fits the model via WLSMV; extracts EBM factor scores. Output: <code>fs_esem.dta</code> .
04_CFA_bifactor.R	Estimates bifactor and unidimensional confirmatory factor models on EYFSP items using WLSMV. Extracts and exports factor scores. Outputs: <code>fs_bifactor.dta</code> , <code>fs_unifactor.dta</code> .
05_PCA_absences.do	Constructs composite indices of absenteeism and social care difficulties using polychoric PCA (categorical variables) and standard PCA (continuous variables). Output: <code>pca_scores.dta</code> .

analysis/

File	Purpose
01_descriptives.do	Loads the analysis sample; merges factor scores; standardises and sign-reverses skill measures; produces summary statistics tables for the full sample and vulnerable subgroups (SEND, CIN/CLA, excluded).
02_eyfsp_differences.do	Plots the percentage-point difference in the distribution of EYFSP point scales between offenders and non-offenders, by subscale.
03_regressions.do	Main regression analysis. Estimates the effects of EYFSP cognitive (<code>cogn_esem</code>) and socio-emotional (<code>semo_esem</code>) factor scores on SEND provision, Child in Need status, absenteeism, exclusions, KS2 attainment, and adolescent offending (any, by type, and count). Appendices include sibling fixed effects, bifactor robustness, and IPW sample-selection correction.
04_ipw.do	Estimates inverse probability weights to correct for sample selection into the EYFSP analytical sample. Logit and Mundlak-adjusted logit models; stabilised and winsorised weights. Output: <code>ipw.dta</code> .
05a_main_mediation.do	Bootstrap wrapper for the mediation analysis. Resamples clusters (schools) with replacement across 250 iterations; calls <code>05b_mediation_boot.do</code> on each iteration. Output: <code>coeff.dta</code> .
05b_mediation_boot.do	Inner loop of the bootstrap procedure. Estimates Level 1 and Level 2 indirect effects of cognitive and socio-emotional skills on offending via school pathways (absences, exclusions, KS2) and vulnerability mediators (SEN, CIN). Stores coefficients on each iteration.

Variable Naming Conventions

Prefix / suffix	Meaning
<code>cogn_esem</code>	Cognitive factor score (ESEM); standardised, sign-reversed
<code>semo_esem</code>	Socio-emotional factor score (ESEM); standardised, sign-reversed
<code>*_bfs</code>	Bifactor model scores (<code>gend</code> , <code>cogn</code> , <code>semo</code>)
<code>sen*</code>	Special Educational Needs indicators (any; by type)
<code>cin</code>	Child in Need indicator
<code>cla</code>	Children Looked After indicator
<code>ks2_*</code>	Key Stage 2 attainment variables
<code>absences_score</code>	Absenteeism composite (continuous PCA)
<code>care_diff_alt</code>	Social care difficulties composite (continuous PCA)
<code>id_dfe</code>	Anonymised pupil identifier (DfE)
<code>id_moj</code>	Anonymised pupil identifier (MoJ)
<code>ipw</code>	Inverse probability weight (baseline logit, stabilised)
<code>ipw_school</code>	Inverse probability weight (Mundlak-adjusted logit)

Notes on Reproducibility

- All file paths use the placeholder `file_path` and must be updated before running.
- Scripts must be run in the order shown within each subfolder. `clean_data/` must be completed before `factorization/` or `analysis/`, as `clean_data.dta` is required by both.
- `clean_data/01` (`eyfsp_clean.dta`) must be run before `clean_data/02`, as the census cleaning script merges on the EYFSP sample to restrict to the analytical cohort.
- The ESEM R script (`factorization/03`) and CFA R script (`factorization/04`) must be run before the main regressions, as the analysis scripts require `fs_esem.dta` and `fs_bifactor.dta`.
- The PCA script (`factorization/05`) must be run before `analysis/01` and `analysis/03`, as it produces `pca_scores.dta`.
- Factor scores (`cogn_esem`, `semo_esem`) are standardised and sign-reversed so that higher values indicate greater developmental difficulty, consistent with the paper's framing.
- The sibling group identifier (`groupid_sib`) is taken from the 2016 census wave only, as the identifier is not stable across waves. All three cohorts are present in the 2016 wave.
- The mediation bootstrap (`analysis/05a`) calls `analysis/05b` as an included do-file on each iteration. Both files must be in the same directory.
- Replication code is also available at: https://github.com/paulgah/Socio-Emotional_Offending

Contact

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